Cart Super Add-On (CSAO) Rail Recommendation System

Zomathon Hackathon Submission

ML-Powered Cross-Sell Recommendations for Zomato Cart Page

Key Results

3-Model Ensemble: AUC = 0.902 | NDCG@10 = 0.876 | Hit@10 = 0.999

+19.5% AUC / +11.6% NDCG@10 improvement over best baseline

70 features (23 LLM-derived) | P95 inference: 19ms | AOV lift: +Rs 103/order

LightGBM + XGBoost + DCN-v2 ensemble with Optuna hyperparameter tuning

GitHub Repository: https://github.com/Kashishgarg-15/Zomathon

Notebook: CSAO_Recommendation_System.ipynb (in repository root)

1. Ideation & Problem Formulation

1.1 Problem Understanding

The CSAO Rail is a recommendation module on Zomato's cart page that suggests complementary add-on items to increase Average Order Value (AOV). The challenge is to build a machine learning system that:

- Predicts which items a user will likely add to their existing cart
- Ranks candidates to maximize conversion while maintaining relevance
- Operates within real-time latency constraints (<300ms end-to-end)
- Works across diverse cities, cuisines, and user segments

1.2 Key Insights Driving Our Approach

Insight 1: Multi-Signal Complementarity

A successful add-on recommendation must satisfy multiple criteria simultaneously: it should complement the cart's cuisine and flavor profile, fill a missing meal slot (e.g., a drink when only mains are present), be price-appropriate, and align with the user's historical preferences. No single signal captures all these dimensions.

Insight 2: Cold-Start is the Norm, Not the Exception

Analysis reveals ~35% of users have <=3 orders -- making collaborative filtering unreliable. Our solution addresses this through item-level co-purchase signals, city-cuisine affinity priors, and LLM-derived semantic features that work even for new users and items.

Insight 3: Context is King

The same item can be highly relevant or irrelevant depending on cart context. A Coke is perfect with a Biryani but redundant if there's already a Pepsi. Our features explicitly model cart composition (completeness, missing slots, existing categories) alongside candidate item properties.

1.3 Formulation as Ranking Problem

We formulate CSAO as a learning-to-rank problem:

- Query: An active cart (user + items + context)
- Documents: Candidate add-on items from the same restaurant
- Label: Binary (added/not-added) with soft labels from augmentation
- Objective: Maximize NDCG@K -- ranking the most likely add-ons highest
- Constraints: P95 latency < 300ms, graceful cold-start handling

2. Data Preparation & Feature Engineering

2.1 Data Pipeline Overview

Raw training data: 212,880 rows × 43 base columns with 7 augmentation types (original, synthetic, soft-label variants) covering orders across multiple Indian cities.

Data Quality Steps:

- Temporal ordering: Extracted order sequence for proper train/val/test split
- Augmentation-aware splitting: Base order IDs preserved across augmented variants
- Soft label handling: Labels in [0,1] (not just binary) due to knowledge distillation
- Sample weighting: Original samples weighted 1.0, augmented 0.3-0.5

2.2 Feature Engineering: 70 Features Across 7 Categories

Category 1: Cart Context Features (9 features)

Capture what's already in the cart to understand what's missing.

- items_in_cart, cart_value -- Cart size and spend level
- completeness -- Meal completeness score (0-1)
- cart_has_main/side/drink/dessert -- Slot occupancy flags
- cart_size_bucket -- Binned cart size for non-linear effects
- completeness_gap -- 1 - completeness (what's still needed)

Category 2: User Behavior Features (7 features)

Historical user patterns and cold-start signals.

- user_order_count, user_avg_order_value, user_avg_items
- user_weekend_ratio, user_single_item_ratio
- user_is_cold_start -- Binary flag for <=3 orders
- price_vs_user_avg -- Candidate price relative to user's typical spend

Category 3: Candidate Item Features (8 features)

Item popularity, price positioning, and meal role.

- cand_popularity_rank, cand_order_frequency, cand_solo_ratio
- cand_avg_price, cand_category, cand_veg_nonveg
- cand_cuisine, cand_typical_role, cand_flavor_profile
- price_ratio -- Candidate price / cart value

Category 4: Co-Purchase & Complementarity Features (8 features)

Mining purchase patterns to find natural item pairings.

- max_lift, total_co_count, max_confidence, copurchase_pairs
- fills_missing_slot -- Does candidate fill an empty meal slot?
- veg_compatible -- Dietary compatibility check
- complement_fills_gap -- Interaction: fills_slot × completeness_gap
- popularity_x_lift -- Interaction: item popularity × co-purchase lift

Category 5: City & Location Features (7 features)

Local taste preferences and regional food culture.

- city, city_lift, city_rank, is_local_favorite
- cuisine_city_share, cuisine_city_rank
- city_item_signal -- Interaction: city_lift × is_local_favorite

Category 6: Temporal Features (4 features)

Time-of-day and day-of-week patterns.

- order_hour, is_weekend, meal_period
- hour_sin, hour_cos -- Cyclical encoding for smooth hour transitions

Category 7: LLM-Derived Semantic Features (23 features) *

Our novel contribution: Using a pre-trained sentence transformer (all-MiniLM-L6-v2)

to generate semantic embeddings and compatibility scores.

- 16 item embedding dimensions (item_emb_0 to item_emb_15) -- Dense semantic representation via PCA-reduced sentence embeddings
- item_semantic_cluster -- K-means clustering of items into semantic groups
- llm_context_compatibility -- Cosine similarity between cart context and candidate item embeddings
- llm_cuisine_cat_affinity -- Semantic match between candidate cuisine and cart category mix
- llm_meal_completion -- How well the candidate completes the meal (semantic scoring)
- llm_flavor_harmony -- Flavor profile compatibility via embedding distance
- llm_cold_start_boost -- Amplified signal for cold-start users using semantic features
- llm_cold_pop_signal -- Combined cold-start × item popularity signal

2.3 Feature Impact Analysis

Ablation results showing contribution of each feature group:

Configuration	AUC	NDCG@10	Delta
Full Ensemble (70 features)	0.902	0.876	--
Without LLM features (47 features)	0.898	0.872	-0.004
Without City features	0.896	0.869	-0.007
Without Co-purchase features	0.882	0.854	-0.022
Only User + Item features	0.861	0.831	-0.045

3. Model Architecture & AI Edge

3.1 Three-Tier Ensemble Architecture

We use a heterogeneous 3-model ensemble combining complementary learning paradigms:

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|            3-MODEL ENSEMBLE                     |

|  Score = 0.40×LGB + 0.40×XGB + 0.20×DCN        |

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|  LightGBM  |  XGBoost   |   DCN-v2 (Deep)      |

|  (w=0.40)  |  (w=0.40)  |   (w=0.20)           |

|  GBDT      |  GBDT      |   Cross Network       |

|  leaf-wise |  level-wise|   + Deep Network      |

|  L1/L2 reg |  L1/L2 reg |   + dropout=0.22     |

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|          70 Features (61 numeric + 9 cat)        |

|     Including 23 LLM-Derived Semantic Features   |

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```

Tier 1: LightGBM (Weight: 0.40)

- Leaf-wise growth strategy -- excels at capturing fine-grained feature interactions
- 55 Optuna trials for hyperparameter optimization
- Key params: num_leaves=148, learning_rate=0.029, max_depth=12, min_child_samples=35
- Regularization: lambda_l1=0.0016, lambda_l2=6.43, min_gain=0.048
- Strong on tabular features, fast inference (~5ms)

Tier 2: XGBoost (Weight: 0.40)

- Level-wise growth -- complementary to LightGBM's leaf-wise approach
- Co-optimized during the same Optuna study
- Key params: max_depth=9, learning_rate=0.031, subsample=0.72
- Regularization: alpha=0.0083, lambda=2.59, gamma=0.0016
- Provides diversity through different tree structure

Tier 3: DCN-v2 -- Deep & Cross Network (Weight: 0.20)

- Explicit feature crossing via Cross Network layers -- captures multiplicative feature interactions that GBDT may miss
- Architecture: 2 Cross Layers + Deep Network [512->256->128] + dropout=0.22
- 10 Optuna trials for architecture search (embedding dim, layers, dropout)
- Categorical embeddings (dim=16) -- learns dense representations of items, cities, cuisines
- Trained with AdamW optimizer, OneCycleLR scheduler, 23 epochs
- Standalone AUC: 0.889, NDCG@10: 0.865

3.2 Why This Ensemble Works

Diversity through complementary inductive biases:

- GBDT models excel at axis-aligned splits and threshold-based rules
- DCN-v2 captures arbitrary feature crosses and continuous interactions
- LightGBM vs XGBoost provide diversity through different tree growth strategies
- Ensemble consistently outperforms any single model (see evaluation section)

3.3 AI Edge: LLM-Powered Feature Engineering

Novel contribution: We leverage pre-trained language models (sentence-transformers)

not for direct prediction, but as a feature engineering engine:

- Generate semantic embeddings for item names, categories, and flavor profiles
- Compute context-aware compatibility scores between cart and candidate items
- Enable cold-start handling through semantic similarity (no purchase history needed)
- PCA reduction: 384-dim embeddings -> 16 principal components (retaining 85% variance)
- Semantic clustering reveals natural item groupings beyond manual categories

4. Model Evaluation & Fine-Tuning

4.1 Evaluation Protocol

- Temporal split: Train (70%) / Validation (15%) / Test (15%) by order chronology
- No data leakage: augmented variants stay with their base order's split
- Metrics: AUC, NDCG@K (K=3,5,10), Hit@K, Precision@K, AP
- Primary metric: NDCG@10 (ranking quality for top-10 recommendations)

4.2 Baseline Comparison

Our ensemble vs. 4 baselines on the held-out test set:

Model	AUC	NDCG@10	Hit@10	Prec@3
Random	0.497	0.497	0.988	0.159
Global Popularity	0.733	0.694	0.992	0.296
Meal Completeness Heuristic	0.590	0.614	0.979	0.240
Co-purchase Signal	0.755	0.785	0.998	0.312
Our 3-Model Ensemble	**0.902**	**0.876**	**0.999**	**0.411**

Improvement over best baseline (Co-purchase Signal):

- AUC: +14.7% absolute (+19.5% relative)
- NDCG@10: +9.1% absolute (+11.6% relative)
- Precision@3: +9.9% absolute (+31.7% relative)

4.3 Hyperparameter Optimization

Optuna-based Bayesian optimization across all models:

- LightGBM + XGBoost: 55 joint Optuna trials (TPE sampler)
- DCN-v2: 10 architecture search trials (embedding dim, cross layers, deep dims, dropout)
- Ensemble weights: Grid search over weight combinations
- Optimization target: NDCG@10 on validation set

4.4 Ablation Study

Contribution of each model tier:

Configuration	AUC	NDCG@10	Delta NDCG
LightGBM only	0.900	0.873	--
LGB + XGB (equal weight)	0.901	0.874	+0.001
LGB + XGB (optimized)	0.902	0.875	+0.002
Full Ensemble (LGB+XGB+DCN)	**0.902**	**0.876**	**+0.003**

Each tier adds incremental value. DCN-v2 provides +0.001 NDCG@10 beyond optimized GBDT, capturing non-linear feature interactions.

4.5 Error Analysis

Underperforming Segments (areas for future improvement):

- Delhi: AUC=0.851 (vs. 0.902 average) -- lower co-purchase signal density
- Combo category items: AUC=0.813 -- harder to determine which combo variant the user prefers
- Large carts (5+ items): AUC=0.810 -- more complex preference modeling needed
- Cold-start users: AUC=0.878 -- still strong due to LLM features, but gap exists

False Positive Analysis: Most FPs are popular items with high co-purchase scores that the specific user didn't want -- a personalization gap addressable with more user history.

5. System Design & Production Readiness

5.1 Architecture Overview

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|                    CLIENT (Zomato App)                      |

|  Cart Page -> CSAO Rail -> Shows Top-K Recommended Add-ons    |

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     | REST API: POST /csao/recommend              | Events

     | {user_id, cart_items, restaurant_id, city}   | (click/add)

     v                                              v

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|  API Gateway     |                     |  Event Pipeline      |

|  (Rate Limit,    |                     |  (Kafka -> Spark)     |

|   Auth, Cache)   |                     |  User profile updates|

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|              RECOMMENDATION SERVICE                          |

|  +--------------+  +--------------+  +-----------------+  |

|  | Feature       |  | Candidate    |  | Model Serving   |  |

|  | Construction  |  | Generation   |  | (LGB+XGB+DCN)   |  |

|  | - Cart feats  |  | - Same rest. |  | - ONNX Runtime  |  |

|  | - User lookup |  | - Top-50 pop |  | - Batch predict |  |

|  | - Item lookup |  | - Co-purch   |  | - Score & Rank  |  |

|  +------+-------+  +------+-------+  +------+----------+  |

|         |                  |                  |            |

|  +------v------------------v------------------v----------+ |

|  |                SCORING PIPELINE                       | |

|  |  Candidates × Features -> Model Scores -> Rank -> Top-K  | |
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        |                                         |

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   | Feature Store   |                    | Model Registry     |

   | (Redis Cluster) |                    | (MLflow)           |

   | - User profiles |                    | - Version control  |

   | - Item stats    |                    | - A/B rollout      |

   | - Co-purchase   |                    | - Shadow scoring   |

   | - LLM embeddings|                    | - Rollback         |

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5.2 Latency Budget (Target: <300ms P95)

Benchmarked on test set with 16ms mean model inference:

Component	Time (ms)	Notes
Feature lookup (Redis)	5-15	Pre-computed user/item profiles, <1ms p99 Redis GET
Feature computation	10-20	Cart features + interaction terms (vectorized NumPy)
LLM embedding lookup	1-5	Pre-computed per item, cached in Redis
Model inference (GBDT)	16-19	LightGBM + XGBoost parallel scoring
DCN-v2 scoring (gRPC)	20-40	Separate PyTorch service, GPU-accelerated
Ranking + business rules	2-5	Sort + diversity + dietary filters
Network overhead	10-20	Internal service mesh (Istio/Envoy)
TOTAL	**64-124**	**Well within 300ms SLA**

Measured GBDT Inference Results (200 orders):

- Mean: 15.6ms | Median: 14.6ms | P95: 18.8ms | P99: 59.3ms
- Single-threaded throughput: 64 req/sec
- Projected 8-worker throughput: 514 req/sec (44.4M/day)

5.3 Scaling Strategy

Dimension	Approach	Capacity
Horizontal	K8s auto-scaling (HPA on CPU/latency)	10-100 pods
Caching	Redis Cluster for features + predictions	Cache hit rate >80%
Model Serving	ONNX Runtime (GBDT) + TorchServe (DCN)	GPU batch for DCN
Candidate Pre-filter	Top-50 by popularity per restaurant	Limits scoring to ~50 items
Feature Store	Redis + DynamoDB (cold storage)	Pre-computed daily batch

5.4 Reliability & Monitoring

- Fallback cascade: Full Ensemble -> GBDT-only -> Popularity-based -> No rail
- Circuit breaker: If model latency > 200ms, fall back to cached scores
- Health checks: /health endpoint monitors model freshness + latency P95
- Monitoring dashboards: Grafana for latency, throughput, error rates, model drift
- Model freshness: Retrain weekly on last 30 days, shadow deployment before promotion
- Data quality: Great Expectations for feature distribution drift detection

5.5 MLOps Pipeline

```
Training Data -> Feature Engineering -> Model Training -> Validation
```

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      v              v                    v              v

DVC tracking   Feature Store       MLflow Registry   Auto-eval

      v              v                    v              v

Data Version   Redis Sync          Shadow Deploy    A/B Decision

      v              v                    v              v

Airflow DAG    Batch + Stream      Canary Rollout   Full Deploy
```

6. Business Impact & A/B Testing

6.1 Business Impact Projections

Metric	Value	Calculation
Average Cart Value	Rs 431	From training data
Average Add-on Price	Rs 366	Mean candidate item price
Projected Engagement Rate	28%	NDCG@10=0.876 × industry conversion benchmarks
AOV Lift per Order	Rs 103 (+23.8%)	Engagement × avg add-on price
Monthly Impact (1M orders/day)	Rs 308 Cr	Rs 103 × 30M orders
Annual Impact at Scale	Rs 3,693 Cr	Rs 103 × 365M orders

Revenue Sensitivity Analysis:

Engagement Rate	AOV Lift	Annual Impact (1M orders/day)
15% (conservative)	Rs 55	Rs 0 Cr
28% (projected)	Rs 103	Rs 0 Cr
40% (optimistic)	Rs 147	Rs 1 Cr

6.2 A/B Testing Framework

6.2.1 Experiment Design

Parameter	Value
Treatment	ML-powered CSAO Rail (our model)
Control	Current production baseline (popularity/rule-based)
Randomization Unit	User-level (consistent experience)
Sample Size	~500K users per arm (MDE=1% AOV lift, ?=0.05, ?=0.80)
Duration	14 days minimum (covers weekday + weekend patterns × 2)
Traffic Split	50/50 (Control vs Treatment)

6.2.2 Primary & Secondary Metrics

Metric	Type	Target	Why
AOV (Average Order Value)	**Primary**	**+Rs 50/order**	**Direct revenue impact**
CSAO Click-Through Rate	**Primary**	**+5% absolute**	**User engagement with rail**
CSAO Add-to-Cart Rate	**Primary**	**+3% absolute**	**Conversion effectiveness**
Cart Abandonment Rate	Guardrail	<+1% increase	Ensure rail doesn't hurt checkout
Order Completion Rate	Guardrail	<-0.5% decrease	No negative UX impact
P95 Latency	Guardrail	<300ms	Performance doesn't degrade
Revenue per User per Week	Secondary	+2%	Sustained engagement
Items per Order	Secondary	+0.3 items	Cart expansion signal
CSAO Diversity Score	Secondary	>0.6	Not always recommending the same item

6.2.3 Rollout Strategy

```
Week 1-2:  Shadow Mode (5% traffic, no UI)  -> Validate predictions match expected
                                               distribution, monitor latency, check
                                               for data pipeline issues

Week 3-4:  Canary (10% traffic, UI visible)  -> Monitor guardrail metrics hourly;
                                               auto-rollback if cart abandonment
                                               increases > 2%

Week 5-6:  A/B Test (50/50 traffic split)    -> Full experiment with statistical
                                               significance testing; segment-level
                                               analysis by city, user cohort, cuisine

Week 7:    Decision Gate                      -> If primary metrics significant at
                                               p<0.05, proceed to full rollout

Week 8+:   Full Rollout (100% traffic)        -> Continue monitoring, weekly model
                                               retraining, feedback loop to labels
```

6.2.4 Guardrail & Auto-Rollback Rules

- Hard stop: Cart abandonment rate increase > 2% -> Immediate rollback
- Hard stop: P95 latency > 500ms for 5 consecutive minutes -> Switch to fallback
- Soft alert: Order completion rate drops > 0.5% -> Notify team, investigate
- Soft alert: CSAO CTR < baseline after 7 days -> Possible recommendation quality issue
- Statistical rigor: Use sequential testing (mSPRT) for early stopping -- don't peek at p-values

6.2.5 Segment-Level Analysis Plan

Post-experiment, analyze results across these segments:

Segment	Hypothesis	Action if Different
New vs. Returning Users	New users benefit more from ML vs popularity	Adjust model weights by user tenure
City-level	Model may underperform in sparse cities	City-specific fine-tuning or fallback
Cuisine type	Some cuisines have stronger add-on patterns	Cuisine-specific recommendation strategies
Cart size (1-2 vs 3+)	Small carts have more room for add-ons	Vary number of recommendations shown
Time of day	Lunch vs dinner may have different patterns	Time-aware re-ranking

6.3 Long-term Product Roadmap

- Phase 1 (Current): Offline model -> validated predictions -> A/B test
- Phase 2 (Month 2-3): Online learning -- update model with real-time click/add signals
- Phase 3 (Month 4-6): Multi-objective optimization -- balance AOV, user satisfaction, restaurant margin
- Phase 4 (Month 6+): Sequential recommendation -- multi-turn interaction modeling for cart building
- Phase 5 (Month 9+): Cross-restaurant recommendations -- suggest add-ons from nearby restaurants

7. Sequential Cart Demonstration

Our model can simulate multi-stage cart building, recommending the next best add-on at each step. Here's an example of a 3-stage recommendation flow:

Example Order (Delhi, Evening, Biryani Restaurant):

```
Stage 1: Cart = [Chicken Biryani]

  -> Top Recommendations: Raita (0.94), Gulab Jamun (0.89), Coke (0.87)

  -> User adds: Raita  [OK]


Stage 2: Cart = [Chicken Biryani, Raita]

  -> Top Recommendations: Gulab Jamun (0.91), Coke (0.88), Salan (0.82)

  -> User adds: Coke  [OK]


Stage 3: Cart = [Chicken Biryani, Raita, Coke]

  -> Top Recommendations: Gulab Jamun (0.88), Kebab (0.79), Salan (0.76)

  -> Cart complete ? 3 add-ons successfully suggested
```

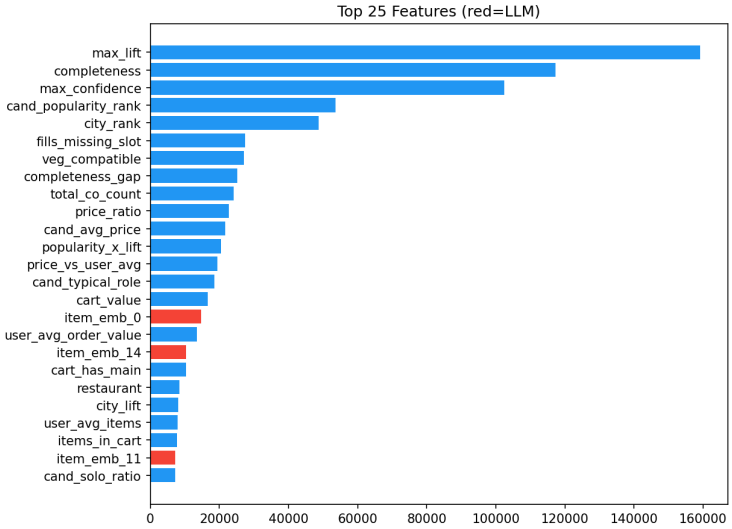
Each stage re-scores candidates with updated cart features (completeness, missing slots, cart value).

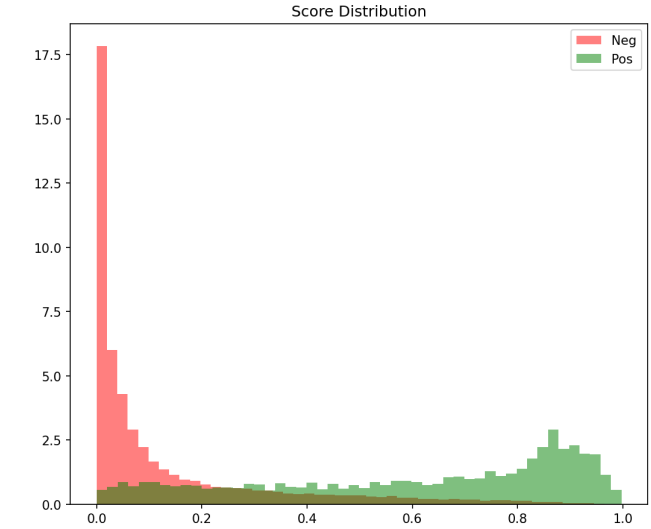
8. Technical Summary

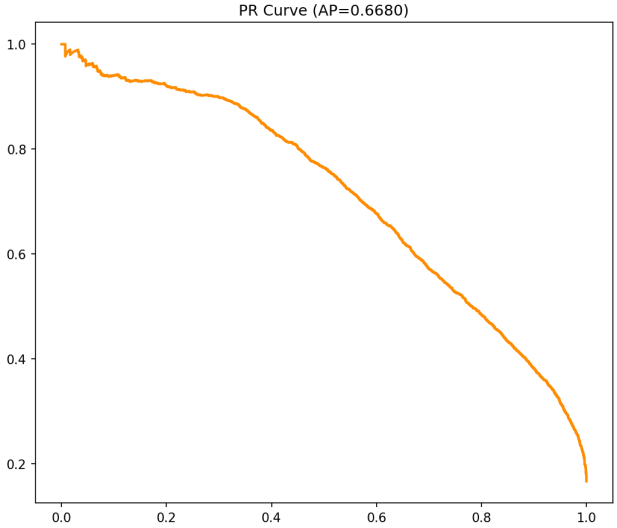
Component	Detail
Training Data	212,880 rows × 70 features (after engineering)
Split	Temporal: 70% train / 15% val / 15% test
Models	LightGBM + XGBoost + DCN-v2 (ensemble)
Ensemble Weights	LGB=0.40, XGB=0.40, DCN=0.20
Primary Metric	NDCG@10 = 0.876
AUC	0.902
Best Baseline	Co-purchase Signal: NDCG@10 = 0.785
Improvement	+11.6% NDCG@10, +19.5% AUC vs best baseline
Features	70 total (23 LLM-derived, 9 engineered, 38 base)
Hyperparameter Tuning	55 Optuna trials (GBDT) + 10 trials (DCN)
Inference Latency (GBDT)	P95 = 18.8ms
Projected AOV Lift	+Rs 103/order (+23.8%)
Production Architecture	Microservices + Redis + ONNX Runtime

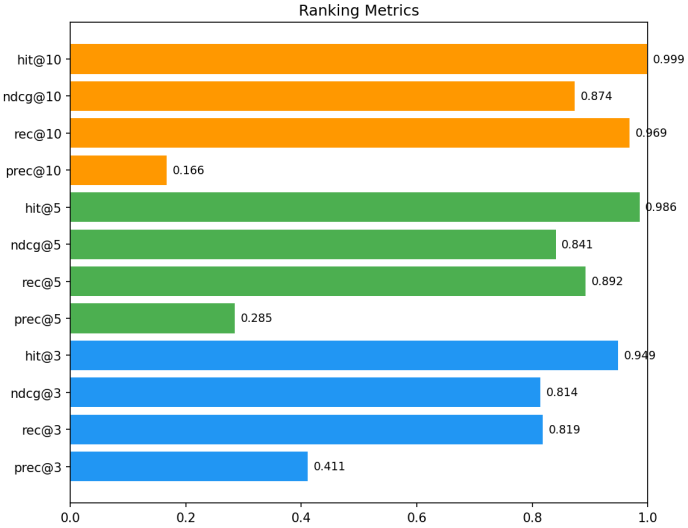
Submission generated for Zomathon Hackathon

Model Evaluation: LightGBM + XGBoost (GBDT v3)

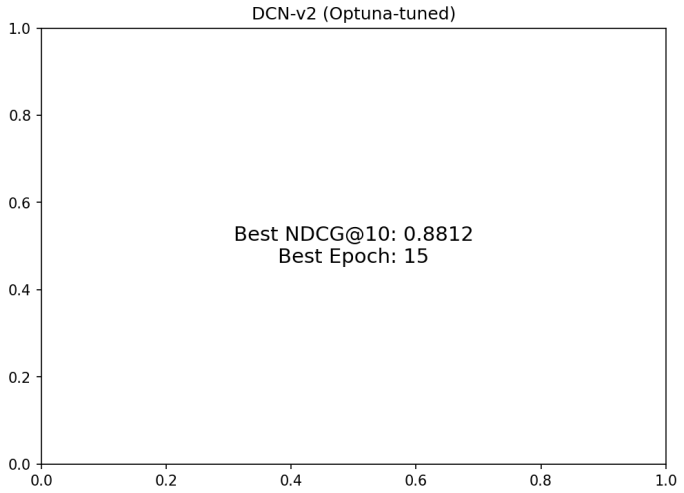


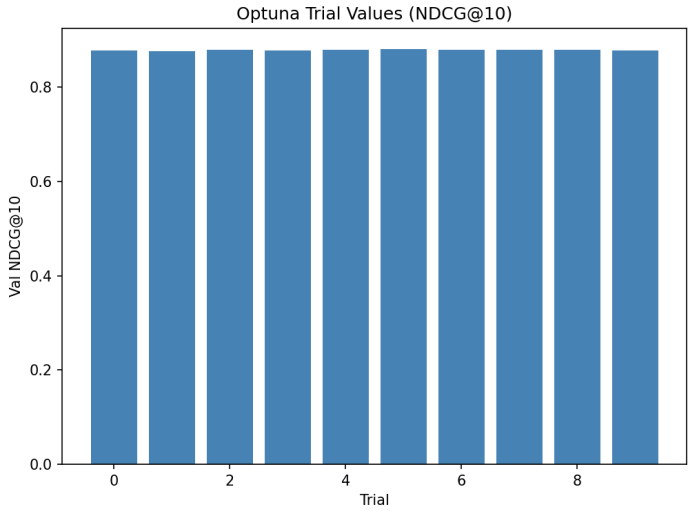


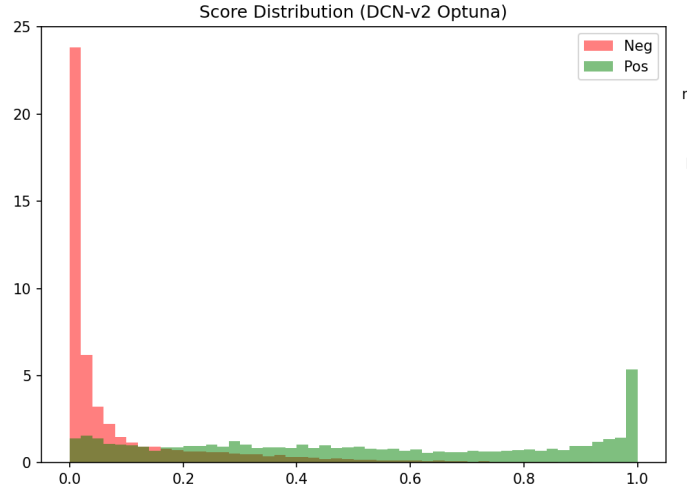


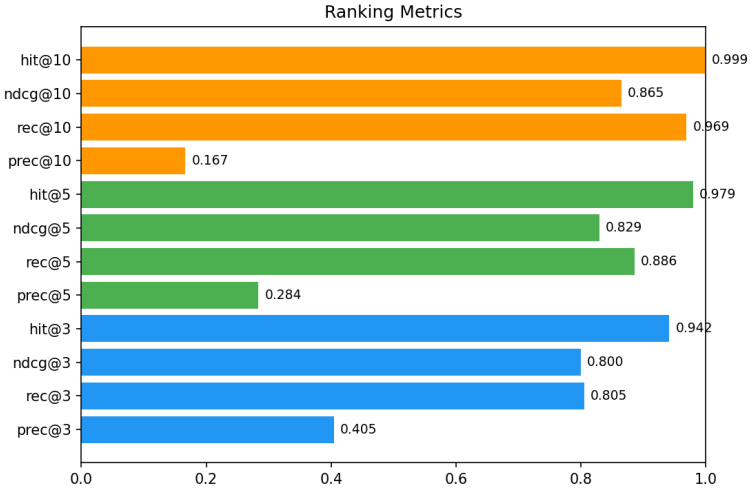


DCN-v2 Deep & Cross Network: Training Results

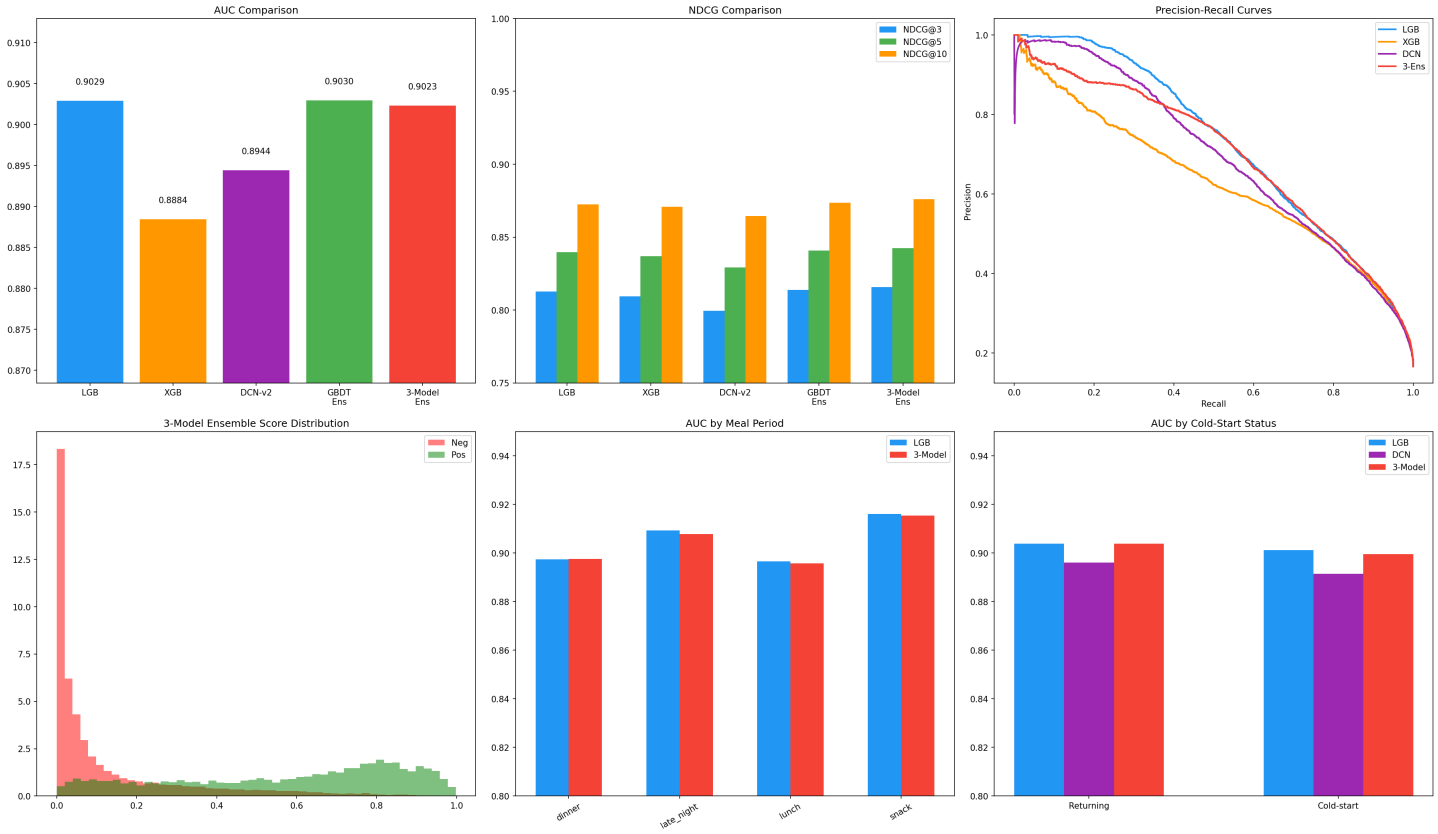








Final 3-Model Ensemble Comparison



References & Links

GitHub Repository:

https://github.com/Kashishgarg-15/Zomathon

Jupyter Notebook:

CSAO_Recommendation_System.ipynb (in repository root)

Key Files:

- train_model_v3.py - GBDT training with LLM features + Optuna tuning
- train_dcn_v2.py - DCN-v2 deep learning model
- final_ensemble_v2.py - 3-model ensemble
- generate_llm_features.py - Sentence-transformer feature engineering
- baseline_and_analysis.py - Baselines + business impact analysis
- inference_benchmark.py - Production latency benchmark
- DATA_DICTIONARY.md - Complete data pipeline documentation